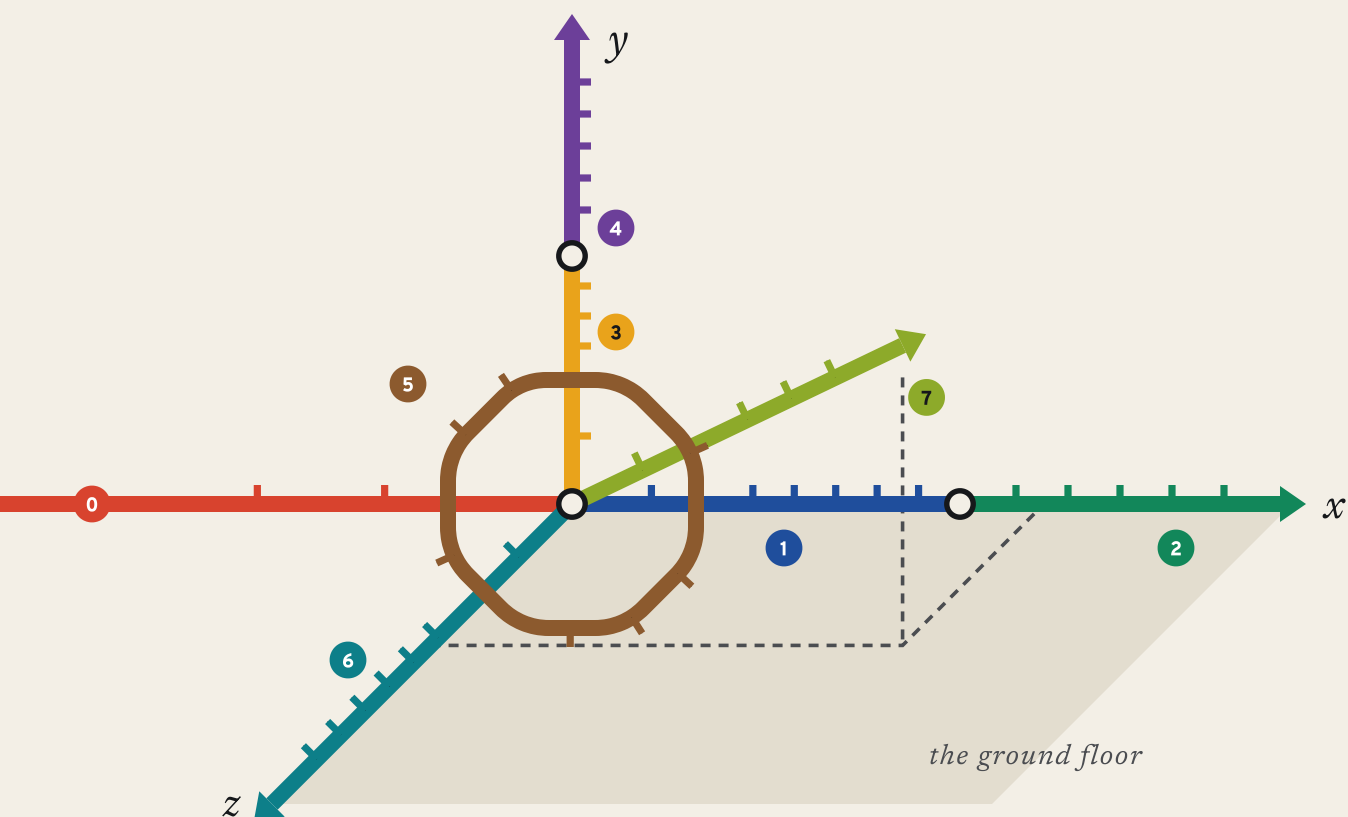


# Mathematical Foundations for Machine Learning

*The linear algebra, probability, and calculus that make ML work.*



|   |                                     |       |   |                                        |       |
|---|-------------------------------------|-------|---|----------------------------------------|-------|
| 0 | WHY MATH FOR MACHINE LEARNING       | 01-03 | 1 | LINEAR ALGEBRA AS TRANSFORMATIONS      | 04-10 |
| 2 | DETERMINANTS AND THE SHAPE OF SPACE | 11-15 | 3 | PROBABILITY AND STATISTICS FOUNDATIONS | 16-20 |
| 4 | PROBABILITY DISTRIBUTIONS           | 21-25 | 5 | PYTHON AND NUMERICAL COMPUTING         | 26-32 |
| 6 | CALCULUS AND THE ENGINE OF LEARNING | 33-39 | 7 | PUTTING IT TOGETHER                    | 40-43 |